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Saveetha Institute of Medical And Technical Sciences
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Security Attacks, Threats, Risks and Their Impact on Security

CSA5115-CRYPTOGRAPHY NETWORK SECURITY

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ABSTRACT



- Information security protects data and systems from unauthorized access, modification, and destruction.
- Cyber threats, attacks, and risks are major challenges that can compromise information security.
- This presentation explains the concepts of security threats, attacks, and risks with relevant examples.
- It discusses how these security issues affect the CIA Triad—Confidentiality, Integrity, and Availability.
- The presentation also highlights common preventive measures used to reduce cyber risks and improve system security.



INTRODUCTION



❖ What is Information Security?

Information Security is the practice of protecting information and computer systems from unauthorized access, misuse, modification, or destruction.

❖ Security Goals (CIA Triad)

Confidentiality – Ensures that information is accessible only to authorized users.

Integrity – Ensures that data remains accurate and is not modified without authorization.

Availability – Ensures that information and services are available whenever required.

Example:

ATM PIN → Confidentiality

Bank Balance → Integrity

Online Banking Service → Availability





SECURITY THREATS, ATTACKS AND RISKS



❖ Security Threat

A threat is a **potential cause of harm** that can exploit a system's weakness.

Examples:

- Hackers
- Insider misuse
- Natural disasters

❖ Security Attack

An attack is an **actual attempt** to damage, steal, or modify information.

Examples:

- Phishing
- Password cracking

❖ Security Risk

A risk is the **possibility of loss** when a threat exploits a vulnerability.

Formula:

$$\text{Risk} = \text{Threat} \times \text{Vulnerability} \times \text{Impact}$$





TYPES OF SECURITY ATTACKS



❖ Passive Attack

- Monitors data without changing it.
- Difficult to detect.
- Affects **Confidentiality**.

Examples:

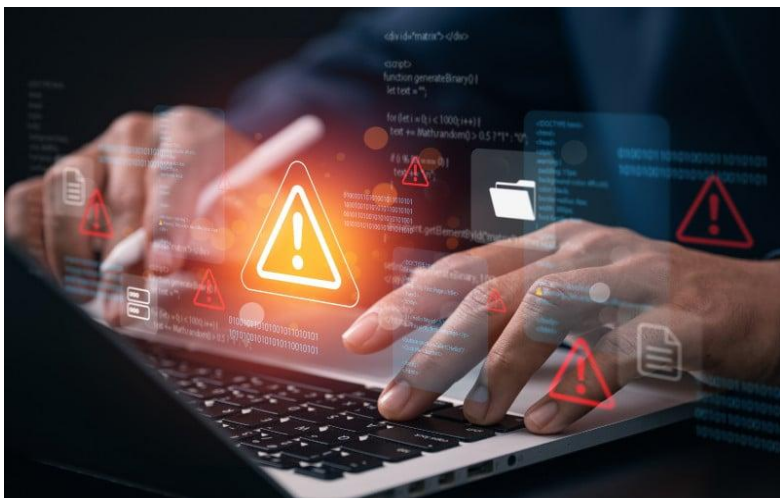
- Eavesdropping
- Traffic Analysis

❖ Active Attack

- Modifies or disrupts data.
- Easier to detect.
- Affects **Integrity and Availability**.

Examples:

- Data Modification
- Replay Attack
- Denial of Service (DoS)

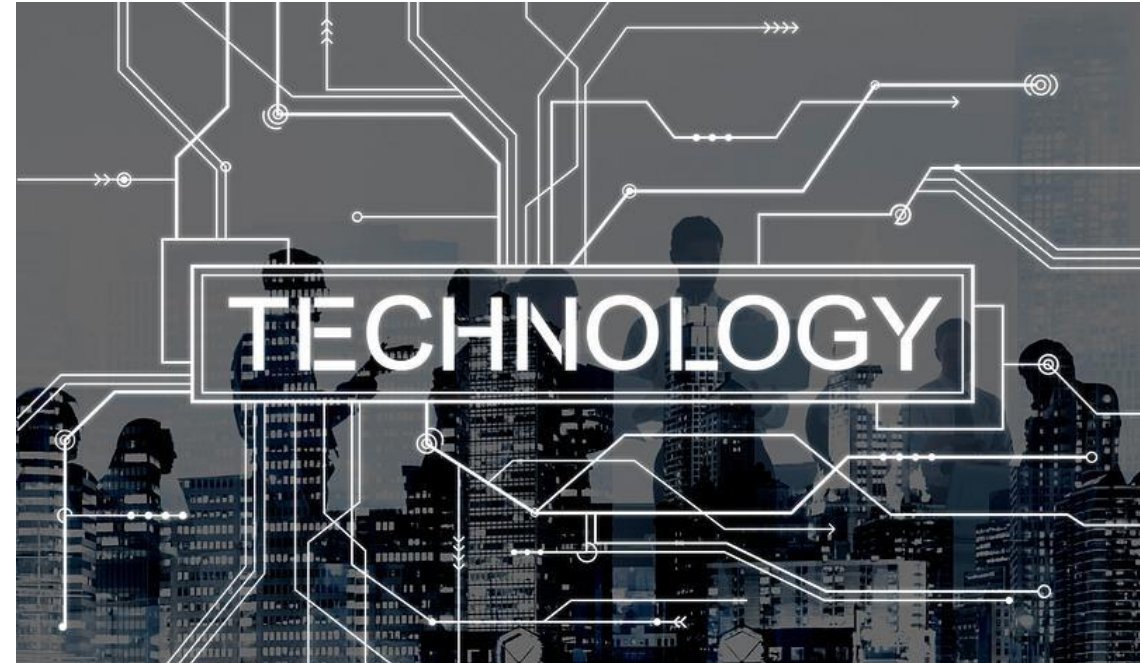




TECHNOLOGIES USED



- Cryptography
- Information Security
- Cybersecurity
- Encryption
- Firewalls
- Multi-Factor Authentication (MFA)
- Intrusion Detection System (IDS)
- Antivirus Software





CONCLUSION



- Information security protects valuable data.
- Threats may become attacks if vulnerabilities exist.
- Risks represent the possible damage caused by attacks.
- The CIA Triad is the foundation of information security.
- Understanding attacks and risks helps organizations build stronger security systems.



thank
you